Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/15/2026
PI Name:	Wang, Zheng Ph.D.		
Project Title:	Harnessing Extremophilic Black Yeasts to Understand the Adaptive Strategies of Eukaryotes to Mars-Like Conditions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	United States Naval Research Laboratory	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	07/23/2021	**End Date:**	03/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:	Balloon Program Office		
Flight Assignment:	NOTE: End date changed to 03/31/2026 per Y. Zhang/KSC (Ed., 5/30/25). NOTE: End date changed to 03/31/2025 per Y. Zhang/KSC (Ed., 5/30/24).		
Key Personnel Changes/Previous PI:	Dr. Tiffany Hennessa is serving as the Co-Investigator. Dr. Hennessa's expertise is plant biology and molecular biology.		
COI Name (Institution):	Hennessa, Tiffany (US Naval Research Laboratory)		
Grant/Contract No.:	NNK20OB08A IAA		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Fungi colonize nearly every ecosystem, including manmade structures. Species that produce airborne spores (i.e., molds) have a ubiquitous global distribution and are therefore the most commonly isolated. Yeasts, the premier model organisms for cell biology, are more restricted in distribution and diversity. Embodying characteristics of both are the black (melanized) fungi; they are hardy and thrive in a wide range of environments, but they do not produce airborne spores, so they can be studied in a restricted laboratory environment. Through natural adaptation to harsh conditions, black fungi have developed tools that situate them to both benefit (as bioremediators of hazardous waste and producers of valuable biomolecules) and threaten (as causes of disease and corrosion of materials) spaceflight and other ventures. These characteristics have motivated increased study of melanized yeasts in the last decade. We propose, here, to further develop these organisms for understanding microbial adaptation in the extreme conditions of space. This project will use state-of-the-art genetic tools to understand how spaceflight affects the survival strategies of black fungi to the conditions experienced on high-altitude Antarctic balloon flights, using species that are rationally chosen based on evolutionary relationships. The objectives are to: 1) understand how high-altitude affects survival of desiccated black yeast populations; 2) identify signatures of spaceflight on biomolecules (DNA, protein) in these organisms; and 3) analyze the effect of high altitude on the recovery of desiccated black yeast cultures, in order to identify processes related to the survival and adaptation of higher organisms (eukaryotes) to these environments. Biological samples will be produced in strictly resource-limited conditions to simulate the relevant, natural growth environments of melanized yeasts. Therefore, while the completion of this project will greatly increase our understanding of how fungi adapt to space-like conditions, an absolute minimal use of personnel and equipment time will be involved.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Antarctic Balloon piggyback opportunity will fly above the Antarctic and experience Mars-analog conditions, with a unique radiological profile. It also allows for a passive biological cargo to be flown as long as it maintains viability over an extended time through intense UV and ionizing radiation. To take advantage of this mission and produce the most valuable biological data possible, we are using the fungus Aspergillus niger to examine how two important cellular mechanisms, melanization and DNA repair (conserved throughout higher organisms) can assist in survival, protection, and adaptation to environments similar to Mars. This opportunity will yield results that greatly contribute to our knowledge of establishing life in Mars but also poses logistical hurdles that we will address in this report.
Task Progress:	On May 20, 2025, flight samples were returned to the U.S. Naval Research Laboratory (NRL) and ground control samples were returned on May 29, 2025. Following the return of the FLEX-Box and recovery of the fungal samples, we initiated a series of post-flight analyses to assess the effects of the balloon flight environment on spore viability and genomic stability. First, we conducted germination assays to evaluate the ability of the fungal spores to grow following exposure. In parallel, we performed imaging studies using Scanning Electron Microscopy (SEM) to assess any morphological changes to the spores' surface structures. Finally, genomic DNA was extracted from the recovered spores and submitted for sequencing to identify any single nucleotide polymorphisms (SNPs) or other genetic alterations that may have arisen during flight. These analyses will provide critical insights into the resilience and genomic integrity of fungal spores exposed to high-altitude conditions, supporting broader efforts to understand microbial survival and adaptation in extreme environments. ANALYSIS METHODS Discs were resuspended with phosphate-buffered saline (PBS) and vortexed for an hour at 40° C to dissolve milk film, and then combined with 50% glycerol and stored in -80° C until future use. GERMINATION Germination of spores was performed on resuspended cells and 3 uL was drop plated in triplicate on minimal media and Yeast Peptone Dextrose (YPD) plates and grown overnight at room temperature. The following day, germination was calculated by sectioning the middle portion of the microscope image and about 200 spores were counted from each droplet. Survival rate was determined by calculating the number of germinated spores and dividing by the total number of spores. MORPHOLOGY For SEM imaging, spore suspension were transferred to a clean tube after removal from discs and fixed with 2.5% glutaraldehyde and fully dehydrated with ethanol. Spores were then dried with hexamethyldisilazane (HMDS) that were then coated with gold prior to imaging. Analysis of morphological changes is still in progress. SEQUENCING Spore stocks were streaked onto YPD solid media and a single colony was selected and used for regrowth for genomic DNA isolation. All samples directly exposed to UV (unshielded) did not germinate and suggests that viability was completely compromised. However, spore samples not exposed to direct UV (shielded), germinated and used to inoculate a 60 mm petri dish with liquid YPD and grown for 1 week. After a week of growth, the fungal mass was harvested and flash frozen in liquid nitrogen and ground with a mortar and pestle. DNA isolation was performed using cetyltrimethylammonium bromide (CTAB) as a detergent to aid in cell lysis and to separate DNA from polysaccharides and contaminants. High molecular weight genomic DNA was sent for whole genome sequencing. Samples are currently in progress and single nucleotide polymorphism (SNP) analysis will be performed after data is returned to us.
Bibliography Type:	Description: (Last Updated: 06/06/2023)